IKARUS 2026

DATA QUEST

Technical Quiz Question Bank

KG Reddy College of Engineering & Technology | Data Science Club

Prepared for the 2nd-year team-based competition. Topics follow the event plan: Python, SQL, statistics, NumPy, Pandas, data analysis, and introductory AI/ML concepts.

Suggested Stage 1 Format

- 15 multiple-choice questions + 5 True/False questions per quiz paper.
- Suggested time: 15 minutes. Correct answers carry 80% of the quiz score; speed carries 20%, as stated in the event plan.
- Use one paper per team of 2–3 participants. No mobile phones or outside assistance if restricted by organizers.
- The question bank below contains 40 MCQs and 10 True/False questions. Select 20 questions for the live round or use the ready-to-use Paper A.

Section A — Multiple-Choice Question Bank (40 Questions)

Choose one correct answer for each question.

1. In Python, which data type is immutable?

A. list
B. dictionary
C. tuple
D. set

2. What is the output of len([4, 7, 9])?

A. 2
B. 3
C. 4
D. Error

3. Which Python keyword is used to define a function?

A. func
B. define
C. def
D. function

4. What does the expression 10 // 3 return?

A. 3.33
B. 3
C. 1
D. 0

5. Which structure stores unique values?

A. list
B. tuple
C. set

D. string

6. Which statement handles an exception in Python?

A. catch
B. except
C. error
D. rescue

7. What is the result of bool([])?

A. True
B. False
C. None
D. Error

8. Which library is commonly used for numerical arrays in Python?

A. NumPy
B. Flask
C. Tkinter
D. Pygame

9. Which Pandas structure is two-dimensional?

A. Series
B. DataFrame
C. ndarray only
D. Tuple

10. In Pandas, which method shows the first five rows by default?

A. tail()
B. first()
C. head()
D. start()

11. Which Pandas method reports missing values per column?

A. isnull().sum()
B. missing()
C. nullcount()
D. empty()

12. What does df.shape return?

A. Column names
B. (rows, columns)
C. Data types only
D. Missing values

13. Which NumPy function creates an array?

A. np.array()
B. np.frame()
C. np.table()
D. np.series()

14. What is the mean of 2, 4, 6, 8?

A. 4
B. 5
C. 6
D. 20

15. Which measure is most affected by extreme outliers?

A. Median
B. Mode
C. Mean
D. Quartile rank

16. The median of 3, 5, 9 is:

A. 3
B. 5
C. 6
D. 9

17. Standard deviation measures:

A. Central value
B. Spread of values
C. Number of rows
D. Category names

18. A correlation of +1 indicates:

A. Perfect positive linear relationship
B. No relationship
C. Perfect negative relationship
D. Causation

19. Which chart is usually suitable for comparing values across categories?

A. Bar chart
B. Scatter plot only
C. Map only
D. Histogram only

20. A histogram is primarily used to display:

A. Distribution of numerical data
B. Database relationships
C. Source code
D. Network routes

21. Which SQL command retrieves data from a table?

A. GET
B. SELECT
C. FETCHTABLE
D. OPEN

22. Which SQL clause filters rows before grouping?

A. ORDER BY
B. HAVING
C. WHERE
D. GROUP BY

23. Which SQL clause filters grouped/aggregated results?

A. HAVING
B. WHERE
C. FROM
D. DISTINCT

24. What does COUNT(*) count?

A. Only non-null values in one chosen column

B. All rows
C. Only unique rows
D. Only numeric rows

25. Which SQL keyword removes duplicate rows from query results?

A. UNIQUE
B. DISTINCT
C. SINGLE
D. CLEAN

26. A primary key should be:

A. Duplicate and nullable
B. Unique and non-null
C. Text only
D. Always a date

27. Which JOIN returns only matching rows from both tables?

A. INNER JOIN
B. LEFT JOIN
C. CROSS JOIN
D. FULL JOIN

28. Which SQL aggregate calculates an average?

A. SUM()
B. MEAN()
C. AVG()
D. MID()

29. In a dataset, a row commonly represents:

A. An observation/record
B. A feature name only
C. A chart
D. A database engine

30. What is data cleaning?

A. Decorating charts
B. Detecting and handling inaccurate, inconsistent, or missing data
C. Deleting every row
D. Encrypting a laptop

31. What is an outlier?

A. A value unusually far from most observations
B. The column heading
C. A missing table
D. A data type

32. Which is an example of categorical data?

A. Height in cm
B. Temperature
C. Blood group
D. Travel time

33. In supervised learning, training data includes:

A. Inputs and target labels
B. Inputs only, always
C. No examples

D. Only charts

34. Which task predicts a continuous numeric value?

A. Classification
B. Regression
C. Clustering
D. Sorting

35. Which task assigns an email to spam or not spam?

A. Regression
B. Classification
C. Dimensionality reduction
D. Aggregation

36. What is overfitting?

A. Model performs well on training data but poorly on unseen data
B. Model has no parameters
C. Dataset has no columns
D. Data is sorted

37. Why split data into training and testing sets?

A. To test generalization on unseen data
B. To make every row identical
C. To remove all features
D. To guarantee 100% accuracy

38. Which metric is the fraction of correct predictions among all predictions?

A. Recall
B. Accuracy
C. Mean squared error
D. Variance

39. What does AI stand for?

A. Automated Internet
B. Artificial Intelligence
C. Advanced Input
D. Applied Integration

40. Generative AI is designed primarily to:

A. Generate content such as text, images, or code
B. Only store rows in SQL
C. Replace all data cleaning
D. Only sort numbers

Section B — True / False (10 Questions)

1. A Python list can contain values of different data types.

2. NumPy arrays are designed for efficient numerical computation.

3. Pandas DataFrame columns must all have the same data type.

4. SQL WHERE is commonly used to filter rows.

5. Correlation alone proves that one variable causes another.

6. The median is usually less sensitive to extreme outliers than the mean.

7. A classification model predicts a category or class.

8. A test dataset should normally be used repeatedly to tune every model choice.

9. Missing data should always be replaced with zero.

10. Generative AI outputs should be checked for accuracy and possible bias.

Ready-to-Use Paper A — 20 Questions

Suggested time: 15 minutes | 15 MCQs + 5 True/False | Team size: 2–3

1. In Python, which data type is immutable?

A. list
B. dictionary
C. tuple
D. set

2. Which Python keyword is used to define a function?

A. func
B. define
C. def
D. function

3. What does the expression 10 // 3 return?

A. 3.33
B. 3
C. 1
D. 0

4. Which Pandas structure is two-dimensional?

A. Series
B. DataFrame
C. ndarray only
D. Tuple

5. Which Pandas method reports missing values per column?

A. isnull().sum()
B. missing()
C. nullcount()
D. empty()

6. Which NumPy function creates an array?

A. np.array()
B. np.frame()
C. np.table()
D. np.series()

7. What is the mean of 2, 4, 6, 8?

A. 4
B. 5
C. 6
D. 20

8. Standard deviation measures:

A. Central value
B. Spread of values
C. Number of rows
D. Category names

9. Which SQL command retrieves data from a table?

A. GET
B. SELECT
C. FETCHTABLE

D. OPEN

10. Which SQL clause filters grouped/aggregated results?

A. HAVING
B. WHERE
C. FROM
D. DISTINCT

11. Which SQL keyword removes duplicate rows from query results?

A. UNIQUE
B. DISTINCT
C. SINGLE
D. CLEAN

12. Which JOIN returns only matching rows from both tables?

A. INNER JOIN
B. LEFT JOIN
C. CROSS JOIN
D. FULL JOIN

13. Which task predicts a continuous numeric value?

A. Classification
B. Regression
C. Clustering
D. Sorting

14. What is overfitting?

A. Model performs well on training data but poorly on unseen data
B. Model has no parameters
C. Dataset has no columns
D. Data is sorted

15. What does AI stand for?

A. Automated Internet
B. Artificial Intelligence
C. Advanced Input
D. Applied Integration

True / False:

16. A Python list can contain values of different data types.

17. Pandas DataFrame columns must all have the same data type.

18. Correlation alone proves that one variable causes another.

19. A classification model predicts a category or class.

20. Missing data should always be replaced with zero.

Answer Key — Organizer Copy

Keep this section separate from participant question papers.

MCQ Answer Key

Q.No.	Answer	Q.No.	Answer
1	C	21	B
2	B	22	C
3	C	23	A
4	B	24	B
5	C	25	B
6	B	26	B
7	B	27	A
8	A	28	C
9	B	29	A
10	C	30	B
11	A	31	A
12	B	32	C
13	A	33	A
14	B	34	B
15	C	35	B
16	B	36	A
17	B	37	A
18	A	38	B
19	A	39	B
20	A	40	A

MCQ Explanations

1. C — A tuple cannot be changed after creation.

2. B — The list contains three elements.

3. C — Functions are declared with def.

4. B — The // operator performs floor division.

5. C — A set stores distinct elements.

6. B — Python uses try/except blocks.

7. B — An empty list is falsy.

8. A — NumPy provides efficient multidimensional arrays.

9. B — A DataFrame is a table-like 2D structure.

10. C — head() returns the first five rows by default.

11. A — isnull() marks missing values and sum() counts them.

12. B — shape returns a tuple of row and column counts.

13. A — np.array() converts input data to an ndarray.

14. B — The sum 20 divided by 4 is 5.

15. C — The mean uses every numeric value and is sensitive to extremes.
16. B — The middle value is 5.
17. B — It measures dispersion around the mean.
18. A — Correlation +1 means a perfect positive linear association, not necessarily causation.
19. A — Bar charts compare categories.
20. A — Histograms show the distribution of numeric values.
21. B — SELECT queries data.
22. C — WHERE filters individual rows before aggregation.
23. A — HAVING filters groups after GROUP BY.
24. B — COUNT(*) counts rows, including rows containing nulls.
25. B — DISTINCT removes duplicate result rows.
26. B — A primary key uniquely identifies each row and cannot be null.
27. A — INNER JOIN returns rows with matches in both tables.
28. C — AVG() calculates the arithmetic average.
29. A — Rows commonly represent observations and columns represent variables.
30. B — Cleaning improves data quality and consistency.
31. A — An outlier differs substantially from the overall pattern.
32. C — Blood group is a category.
33. A — Supervised models learn from examples with known targets.
34. B — Regression predicts numeric quantities.
35. B — Spam detection is a classification task.
36. A — Overfit models learn training-specific patterns that do not generalize.
37. A — The test set provides an estimate of performance on unseen examples.
38. B — Accuracy is correct predictions divided by total predictions.
39. B — AI means Artificial Intelligence.
40. A — Generative models create new content based on learned patterns.

True / False Answer Key

1. True — Python lists can hold mixed types.
2. True — NumPy supports vectorized numerical operations.
3. False — Different columns can have different dtypes.
4. True — WHERE applies row-level conditions.
5. False — Correlation does not establish causation.
6. True — The median depends on the middle position.
7. True — Classification outputs discrete labels/classes.
8. False — Repeated tuning on test data can cause information leakage; use validation data for tuning.
9. False — The appropriate strategy depends on what is missing and why.
10. True — Generated content can be incorrect or biased.

Suggested Paper A Answer Key

1. C | 2. C | 3. B | 4. B | 5. A

6. A | 7. B | 8. B | 9. B | 10. A

11. B | 12. A | 13. B | 14. A | 15. B

T/F: 16. True | 17. False | 18. False | 19. True | 20. False

Suggested Scoring Method

- Correctness: 80% of the quiz score.
- Speed: 20% of the quiz score. Use a transparent tie-break rule such as earlier valid submission among teams with equal correct scores.
- Recommended: award no negative marks unless announced before the round.
- Print participant papers without the answer-key pages; keep the organizer copy confidential.

Organizer Note

This bank is tailored to IKARUS: DATA QUEST for 2nd-year students and reflects the event plan's listed topics. The document proposes a ready-to-use Stage 1 quiz; confirm final timing, scoring, and permitted resources with the event lead before printing.